

Midhuna Venugopal

Data Analyst

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Data Analyst with 2+ years of experience applying machine learning, deep learning, statistical modeling, and data analysis to solve applied research problems and deliver actionable insights. Proficient in Python (Pandas, NumPy, Scikit-learn, TensorFlow) and R for developing and evaluating analytical solutions, including regression, classification, clustering, neural networks, and predictive modeling. Experienced managing end-to-end analysis workflows, from data validation through reporting, and collaborating with cross-functional research teams to translate complex data into business-relevant insights.

Core Toolkit

LANGUAGES & ENVIRONMENTS

Python (Pandas, NumPy, Scikit-learn, TensorFlow), R, SQL

MODELING & METHODS

Regression, Classification, Clustering, Model Evaluation, Deep Learning, Statistical Modeling, Hypothesis Testing, EDA, Time Series Analysis, NLP

STATISTICAL SOFTWARE

SPSS, Stata, EViews

VISUALIZATION

Power BI, Matplotlib / Seaborn, Excel

Education

M.Sc. Statistics with Data Science

Rajagiri College of Social Sciences

GPA 4.3 / 5.0

B.Sc. Mathematics

St. Paul's College

CGPA 8.39 / 10

Selected Projects

PROJECT 01

Ride Demand Forecasting

Prophet, XGBoost & LSTM on 150K NCR ride-hailing bookings to forecast hourly demand and surface driver-deployment insights.

PROJECT 02

AI Research Knowledge Graph

NLP topic modeling and network analysis over 136K arXiv papers to map research communities and emerging trends.

Ride Demand Forecasting for Indian Ride-Hailing Services

Forecasting hourly ride demand from 150,000 raw booking records across the Delhi-NCR region (Jan–Oct 2024), comparing three modeling paradigms to identify the most reliable approach for operational deployment.

- TIME-SERIES
- PROPHET
- XGBOOST
- LSTM
- FEATURE ENGINEERING

150,000

RAW BOOKINGS ANALYZED

8,760

HOURLY TIME STEPS BUILT

3

FORECASTING MODELS COMPARED

0.831

BEST R² (XGBOOST)

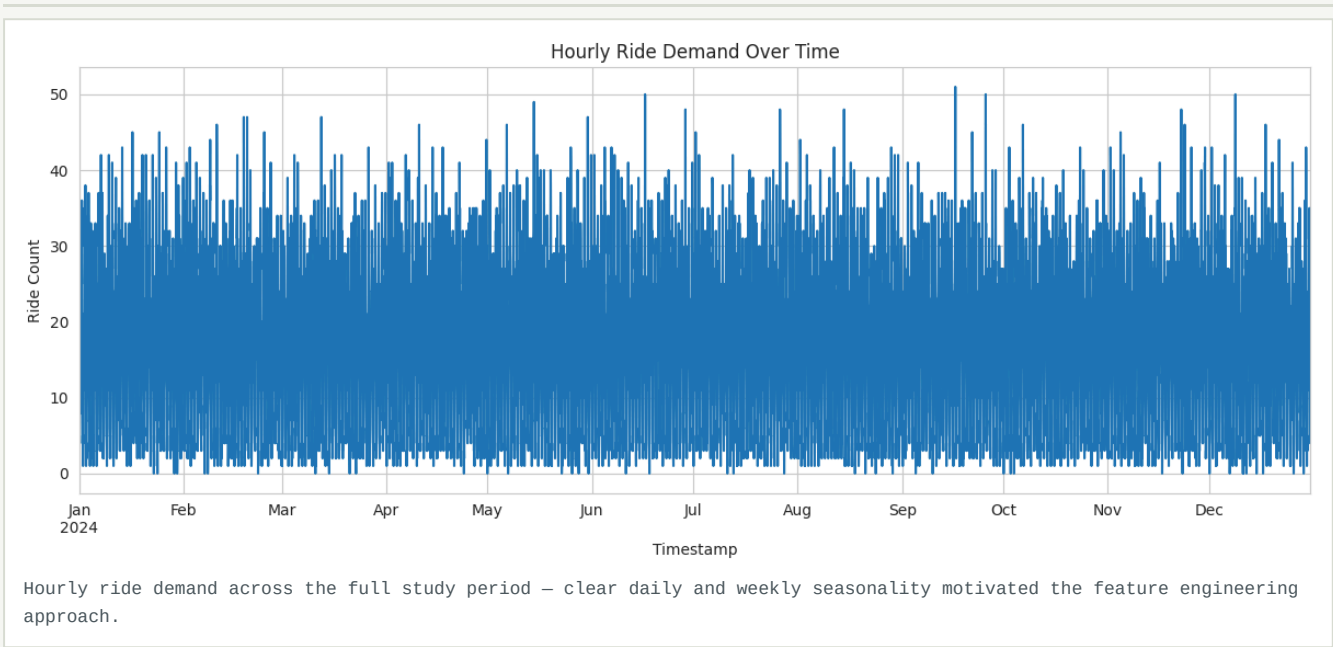
APPROACH

- Data preparation:** parsed timestamps, resolved 1,233 duplicate booking IDs, and separated structural nulls (ride never happened) from true missing values before building a clean hourly ride-count series.
- Feature engineering:** calendar features (hour, weekday, month) plus lag (1h–24h) and rolling mean/std features to give tree-based models temporal memory.
- Modeling:** trained Prophet for trend/seasonality decomposition, XGBoost on the engineered feature set, and an LSTM on scaled 24-hour input sequences — all evaluated on the same chronological hold-out (Dec 17–30).

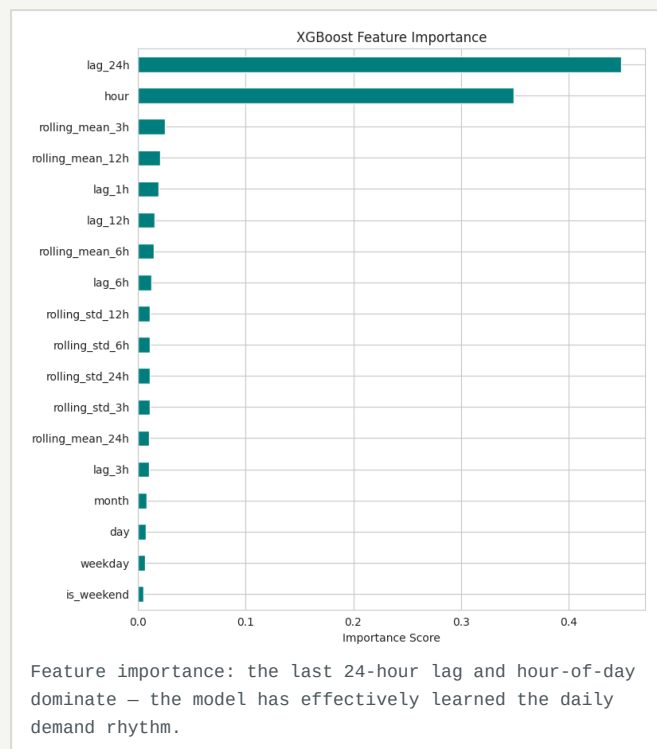
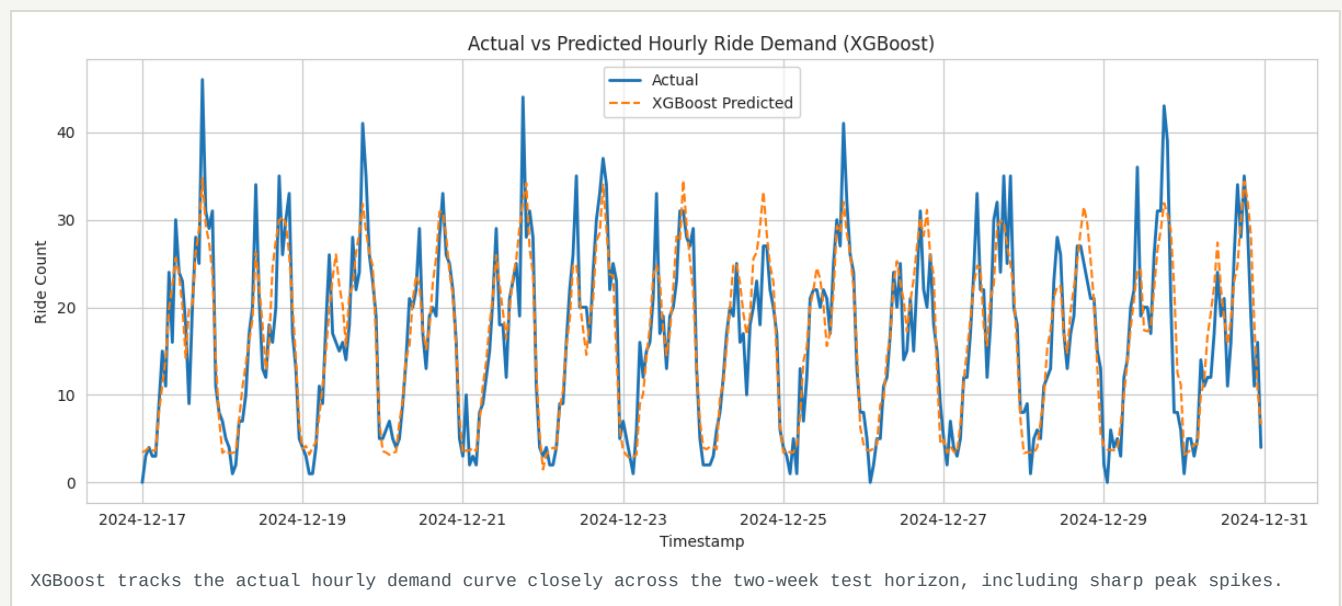
MODEL COMPARISON

MODEL	MAE	RMSE	MAPE (%)	R ²
Prophet	3.260	4.229	31.17	0.822
XGBoost	3.169	4.126	29.75	0.831
LSTM	3.389	4.380	31.55	0.809

VISUAL HIGHLIGHTS



Prediction Quality & Feature Drivers



BUSINESS IMPLICATIONS

- **Driver deployment:** shift schedules can be aligned to forecast peak windows (identified around 5–8 PM) to cut passenger wait times.
- **Dynamic pricing:** peak/off-peak patterns feed directly into surge-pricing rules.
- **Cancellation reduction:** matching driver supply to predicted demand avoids the under/over-supply that drives cancellations.
- **Model choice:** XGBoost's feature-driven approach outperformed both Prophet and LSTM on every metric, making it the recommended model for deployment.

Knowledge Graph-Based Discovery of Emerging Research Trends in Artificial Intelligence Literature

Built an end-to-end NLP and network-analysis pipeline over 136,238 arXiv papers to identify research topics, map relationships between papers, authors and topics, and surface emerging research communities.

NLP

BERTOPIC

NETWORKX

LOUVAIN

RECOMMENDER

136,238

ARXIV PAPERS SCREENED

5,945

GRAPH NODES BUILT

9,000

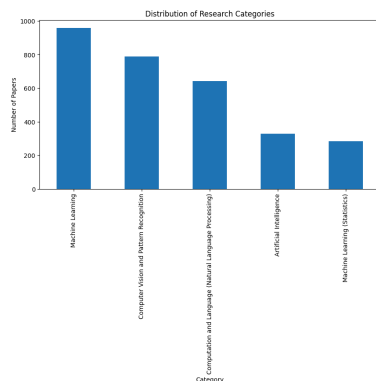
GRAPH EDGES BUILT

15

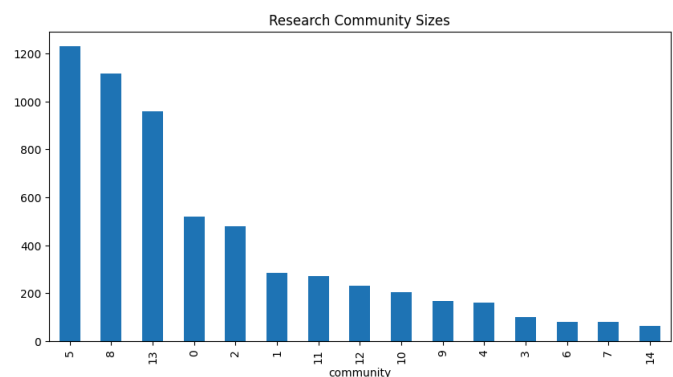
RESEARCH COMMUNITIES FOUND

APPROACH

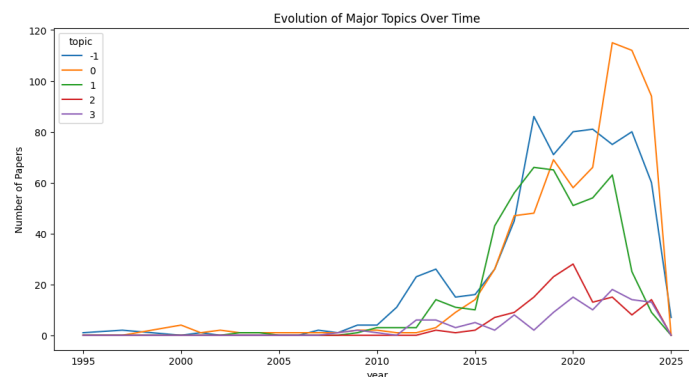
- **Corpus construction:** filtered 136K papers down to 117,661 AI/ML-relevant records across 5 categories, then drew a stratified 3,000-paper sample spanning 1995–2025.
- **Topic modeling:** cleaned abstracts with NLTK, then applied BERTopic to surface 20+ coherent research topics from unstructured text.
- **Knowledge graph:** constructed a heterogeneous graph in NetworkX linking 3,000 papers, 2,919 authors, 21 topics and 5 categories (5,945 nodes / 9,000 edges), then ran Louvain community detection to uncover 15 distinct research communities.
- **Applications:** built a lightweight recommendation engine that surfaces similar papers and adjacent topics given a single input paper.



Machine Learning and Computer Vision dominate the sampled AI literature.



Community sizes from Louvain detection – two large hub communities alongside several smaller specialist clusters.



Evolution of major research topics over time, showing rapid growth in dominant themes post-2016 as deep learning research accelerated.

Thank you for reading — full notebooks available on request.